Introduction

Delivering computing to young learners

Oxford International Primary and Lower Secondary Computing is a complete syllabus for computing education for ages 5–14 (Years 1–9). By following the program of learning set out in this series, teachers can feel reassured that their students have access to the computing skills and understanding that they need for their future education.

Find out more at:
www.oxfordprimary.com/computing.

Structure of the book

This book is divided into six chapters, for Year 5 (ages 9–10).

1. **The nature of technology:** Introduction to computer networks
2. **Digital literacy:** Searching for information on the world wide web
3. **Computational thinking:** How we use loops and other structures in programming
4. **Programming:** Using programming skills to control on-screen actions
5. **Multimedia:** Taking digital photographs
6. **Numbers and data:** Using a spreadsheet to help with realistic business activities

What you will find in each unit

- Introduction: An offline activity and a class discussion help students to start thinking about the topic.
- Lessons: Six lessons guide students through activity-based learning.
- Check what you know: A test and activities allow you to measure students' progress.

What you will find in the lessons

Although each lesson is unique, they have common features: learning outcomes for each lesson are set out at the start; learning content delivers skills and develops understanding.


Activity Every lesson involves a learning activity for the students.


Extra challenge Activities to extend students who are able to do more.



Questions check students' understanding of the lesson.

Additional features

You will also find these features throughout the book:



The word cloud builds vocabulary by identifying key terms from the unit.



Suggestions for creative and artistic work.



Extra tasks that can be taken outside the classroom and into the home.



Advice on using computers responsibly in life.

Glossary Key terms are identified in the text and defined in the glossary at the end.

Assessing student achievement

The final pages in each unit give an opportunity to assess student achievement.

- Developing: This acknowledges the achievement of students who find the content challenging but have made progress.
- Secure: Students have reached the level set out in the programme for their age group. Most should reach this level.
- Extended: This recognises the achievement of students who have developed above-average skills and understanding.

Questions and activities are colour-coded according to achievement level. Self-evaluation advice helps students to check their own progress.

Software to use

We recommend Scratch for writing programs at this age. For other lessons, teachers can use any suitable software, for example: Microsoft Office; Google Drive software; LibreOffice; any web browser.

Source files


You will see this symbol on some of the pages.

This means that there are extra files you can access to help with the learning activities. For example, Scratch programming files and downloadable images.

To access the files, click 'Download resources' at:
www.oxfordprimary.com/computing.

Teacher's Guides

For more on these topics, look at the Teacher's Guide that accompanies this book.

1 The nature of technology: Computer networks

You will learn

- how digital devices can be connected to make networks
- what the internet is and what services it provides
- how the internet helps us work together in the modern world.



A computer is a powerful tool that we use for working and learning. Connecting computers together makes them more powerful and more useful. When computers are connected, we can use them to communicate with each other. We can share files and search for information on the web. We use networks to connect computers together. In this unit you will learn how networks improve the way we live our lives.



Talk about...

Do you think you will use computers in your job when you leave school? Would you like to work in computing when you leave school? Can you think of any jobs that don't use computers?



Learning outcomes: Explain that digital devices can be connected by communication links; Explain what the internet is and some internet services such as the world wide web; Describe some ways the internet helps us work together in the modern world

network network device
local area network
wide area network
internet world wide web
Wi-Fi server hub router



Class activity

Here are two statements about how young people should use the internet.

1 "The internet is dangerous. Young people should not be allowed to use the internet."

2 "Young people should be able to use the internet whenever and wherever they choose."

Discuss the statements in your class. List the arguments for and against each statement.

Can your class write your own statement about how young people should use the internet?



Digital citizen of the future

Whatever you do in your future life, computers and networks will play an important part. The world of computing changes quickly. You will need to keep your skills up to date so you can use technology at work and at home. At work you will go on training courses to learn new skills. At home you will use the internet to learn new skills. As a good digital citizen, you will help people who have fallen behind with their computing skills.



Did you know?

Today, almost every computer is connected to a network. The biggest network in the world is the internet. The internet has the potential to connect computers wherever they are.

